

# Edward Zhuang

☎ (647) 787-4007 | ✉ edwardzhuang2001@gmail.com | 🌐 /edwardzhuang2001 | 📺 /edwardzhuang2001

## Experience

### Software Engineer, Big Data

August 2024 – Present

ROYAL BANK OF CANADA, ANTI-MONEY LAUNDERING (AML)

Toronto, ON

- Enabled AML regulatory coverage for a newly acquired wealth management firm by integrating client and transaction data for 70K+ clients representing \$80B+ in assets into RBC's Hadoop AML data lake, ensuring continuous screening, transaction monitoring, and regulatory compliance post-acquisition
- Implemented Spark SQL ETL pipelines and data validation frameworks to consolidate daily heterogeneous client and transaction datasets, preventing monitoring gaps that could expose the bank to undetected financial crime and regulatory risk
- Modernized payment monitoring logic through SWIFT MX adoption, building a new data model (central SQL view and auxiliary tables) that improved data completeness and reduced false-positive AML alerts by up to 30% compared to legacy formats
- Drove RBC's Hive-to-Snowflake AML platform migration by developing a configurable Java Spring Boot data-flattening service to transform unstructured payment payloads (JSON, Avro) from AWS S3 into Snowflake-ready schemas, improving downstream AML investigator query performance by 10x+
- Built and maintained core components of a Scala/Java data integration platform supporting standardized ingestion and transformation across multiple data formats, enabling delivery of 10+ AML ETL initiatives with consistent controls and reduced onboarding time for new data sources

### Software Engineer Intern, Backend

May 2022 – April 2023

NATIONAL BANK OF CANADA, SECURITIES LENDING

Toronto, ON

- Developed microservices for a securities trading platform in Java Spring Boot, delivering 100+ production features
- Implemented server-side infrastructure to enable the migration of the on-premises, monolithic application to AWS, adopting a microservices architecture using Java Spring Boot, improving scalability and achieving 99.9%+ system uptime
- Designed modules for daily notional value tracking by constructing new SQL tables and implementing Spring Data JPA repositories in Java, generating trade activity insights and reducing manual effort by 20+ hrs/week across 50+ traders

## Projects

### ASLingo

September 2023 – April 2024

CAPSTONE PROJECT

McMaster Capstone Course

- Trained a linear neural network model to recognize the alphabet and common hand signs in American Sign Language (ASL) using PyTorch, based on image recognition built in OpenCV with Python
- Developed a web platform for learning ASL using computer vision featuring real-time practice via video and feedback
- Designed a data processing pipeline backend in Python and a landmark-based hand sign recognition machine learning model

### Opioid Patient Study

September 2021 - April 2022

SCHOOL CLUB PROJECT

McMaster AI Society

- Implemented a support vector machine to classify potential opioid addicts from patients in the Pharmacogenetics of Opioid Substitutions Treatment Response Study, in collaboration with the McMaster Faculty of Health Sciences
- Developed data cleaning Python scripts using pandas, reducing a noisy and inconsistent data set of 3K+ features to a clean data set with < 400 features, minimizing the effects of overfitting

## Skills

**Development** Java, Spring Boot, Python, Flask, Scala, JavaScript (ES6), React, Node.js, HTML, CSS, Shell

**Big Data & ML** SQL, Snowflake, Hadoop, Spark (Spark SQL), Hive, pandas, NumPy, PyTorch

**DevOps & Cloud** Git, AWS, GitHub Actions, Jenkins, Docker, Atlassian

## Education

### McMaster University

September 2019 – April 2024

BACHELOR OF ENGINEERING – SOFTWARE ENGINEERING (CO-OP)

Hamilton, ON

- **Relevant courses:** Data Structures and Algorithms, Concurrent System Design, Large Software Design, Databases, Data Mining